microeconomics_2_L4

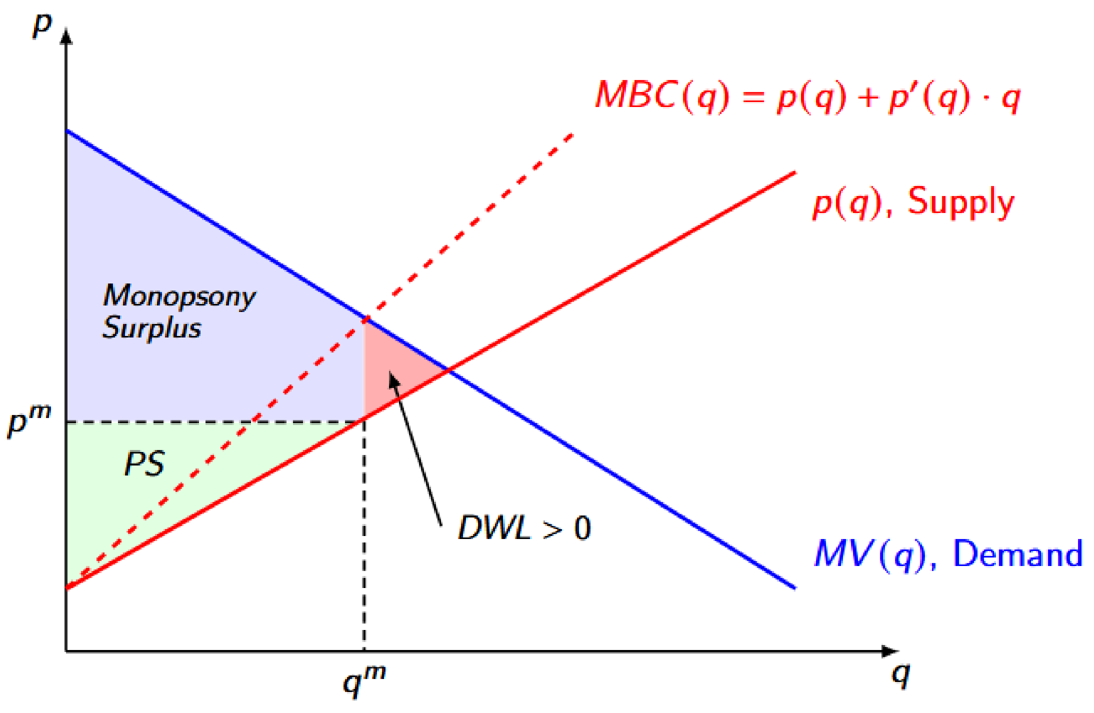
Subject: #microeconomics

Topic: Monopsony

Definition:
A firm is a monopsony if it is the only buyer of a product in a given market.

The characteristics of a monopsony are:

- Value of buying quantity q: V(q) (MV for marginal)
- p(q) is the inverse supply
- Cost of buying q units: $BC(q) = p(p) \cdot q$
- Marginal cost of buying q units: $MBC(q) = p(q) + p'(q) \cdot q$
- Monopsony's profit: $\pi(q) = V(q) - BC(q)$
- Monopsony will choose q^m to satisfy $MV(q^m) = MBC(q^m)$.



Since $MBC(q) > p(q)$, there is always undersupply $q^m < q^*$ and positive DWL.

How to solve problems with multiple consumers?

Step 1: Demand Aggregation:
$D(p)$ tells us the quantity that a consumer wants to buy per unit price p. The inverse demand function

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Topic: Price Discrimination

Definition. Let $T_i(q)$ denote the tariff (payment function) for consumer i purchasing quantity q .

> **Note**
>
> It is assumed that $T_i(0) = 0$ for all i.

Definition. Under linear pricing (no discrimination), the tariff for all consumers i and quantities $q \geq 0$ is given by:

$$T_i(q) = p \cdot q$$

Definition. The marginal revenue for a monopolist using a single per-unit price is:

$$MR(q) = p(q) + q \cdot p'(q), \quad \text{where } p'(q) < 0$$

Definition. Price discrimination aims to **eliminate the negative effect** of $p'(q)$ in the MR equation, allowing a firm to increase sales without reducing the price on all units.

> **Note**
>
> Price discrimination is only feasible if **resale is banned** or too costly.

Types of Price Discrimination

We consider **two settings**:

1. **Unit-Demand Setting**
2. **General Setting**

And **three types** of price discrimination:

1. Third-Degree Price Discrimination

- Linear prices, but **consumer-specific**.
- To buy q units, consumer i pays $T_i(q) = p_i \cdot q$.
- "Consumer i" typically refers to a **group** (location, age, gender, etc.).

2. Second-Degree Price Discrimination

- Only **quantity-specific prices**.
- A single tariff function $T(q)$ is offered to all consumers.
- **Examples:** mobile service plans, bulk discounts.

3. First-Degree (Perfect) Price Discrimination

- Both **consumer-specific** and **quantity-specific**.
- Consumer i pays tariff $T_i(q)$.
- **Challenges:** Often infeasible or illegal; requires detailed consumer data.

Tariffs (Payment Functions)

Definition. A tariff menu is a set of functions $\{T_1(q), T_2(q), T_3(q), \ldots\}$ from which consumers can choose.

Examples of alternative tariff functions:

- $T_1(q) = \begin{cases} 0 & q = 0 \\ 10 & q > 0 \end{cases}$ (Two-part tariff)
- $T_2(q) = \begin{cases} 0 & q = 0 \\ 5 + q & q > 0 \end{cases}$ (Two-part tariff)
- $T_3(q) = \begin{cases} 4q & q \leq 2 \\ 3q & q > 2 \end{cases}$ (Volume discount)

Class Example 1. A rental service uses the following tariff function:

$$T(q) = \begin{cases} 0 & q = 0 \\ 50 + 4q & q > 0 \end{cases}$$

This means a flat fee of **50** plus **4 per unit rented**.

> ⚠ **Warning**
>
> If at any point your calculated consumer payment $T(q) < 0$, then you are doing something wrong.

> **Note**
>
> The firm's problem is to design tariffs to maximize profit subject to consumer participation and self-selection constraints.

Sources:

Additional materials:

Review questions: